

Sid Pallerla

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EDUCATION

University of California, Berkeley

Berkeley, CA

Bachelor of Arts in Computer Science

Aug 2022 - May 2026

Relevant Coursework: Data Structures and Algorithms, Computer Architecture, Efficient Algorithms and Intractable Problems, Structure and Interpretation of Computer Programs, Linear Algebra, Foundations of Data Science, Discrete Mathematics and Probability

EXPERIENCE

Software Engineer Intern

May 2024 – Aug 2024

Redapt Inc

Woodinville, WA

- Improved marketing team's data storage by over **55%** by developing marketing analyst tool
- Enabled the marketing team to produce **25%** more content per month by providing instant access to accurate company information by developing a **retrieval-augmented generation (RAG) model**
- Optimized and deployed **Docker** containers, collaborating with the release team to streamline over **10** successful deployments
- Developed and validated **YAML** files for deployment configurations, services, **Istio** routing, and scaling,
- Partnered with machine learning engineers to update and manage **Azure ML Studio Prompt Flows**, reducing processing time by 30% across key workflows

Undergraduate Academic Intern

Jan 2024 – May 2024

University of California, Berkeley EECS Department

Berkeley, CA

- Assisted with teaching and mentoring students in CS61B, an advanced data structures and algorithms course
- Facilitated weekly lab sessions with **25+** students, providing guidance to students in solving programming problems
- Designed course materials, including **10+** programming assignments, **50+** practice problems, and **3** exams

Undergraduate Researcher

Jan 2024 – May 2024

Berkeley Laboratory for Usable and Experimental Security (BLUES)

Berkeley, CA

- Achieved a **61% improvement** in efficiency for security protocol implementation and testing within the lab by developing a Java mobile application and corresponding SDK
- Analyzed data of over 50 participants in a research study using MATLAB and Python
- Participated in the literary review and contributed to a research paper on computer security protocols

TECHNICAL SKILLS

Languages: Python, Java, Javascript, SQL, C

Frameworks/Libraries: React, Pytorch, Beautiful Soup, Pandas, Flask

Others: GIT, AWS, Linux, Docker, Kubernetes, CSS, HTML

PROJECTS

Marketing Analyst Tool | *Beautiful Soup, React, Pytorch*

- Utilized Beautiful Soup and Google Search API to scrape and aggregate relevant marketing data
- Developed a extractive text summarization model using **PyTorch's HuggingFace Transformers library** to analyze and summarize the scraped data
- Built an interactive frontend using React, allowing marketing analysts to easily input parameters and view results.

Company Information RAG Tool | *Azure ML Studio, Azure AI Search, Azure OpenAI API Service*

- Created a retrieval-augmented generation (RAG) model using Azure **ML Studio's Prompt Flow** and **Azure AI Search** with company data
- Automated the Prompt Flow process by developing code with Azure OpenAI's API
- Deployed as an endpoint for streamlined and scalable access

Reddit Comment Bot | *Python Reddit API Wrapper, Python*

- Developed a Reddit bot using Python and PRAW (Python Reddit API Wrapper) library to automate responses based on specific phrases/triggers
- Automated Reddit interactions, achieving a maximum of **200** comment replies daily